

Practical No. 7 Competitive Code

Name: Rana Chavan

Section: A4 – B2

Roll no. 26

Code:

```
import java.util.ArrayList;

import java.util.Arrays;

class Solution {

    boolean[] visited;

    boolean check(int N, int M, ArrayList<ArrayList<Integer>> Edges) {

        ArrayList<ArrayList<Integer>> adj = new ArrayList<>();

        for (int i = 0; i <= N; i++) {

            adj.add(new ArrayList<>());

        }

        for (ArrayList<Integer> edge : Edges) {

            int u = edge.get(0);

            int v = edge.get(1);

            adj.get(u).add(v);

            adj.get(v).add(u);

        }

        this.visited = new boolean[N + 1];

        for (int i = 1; i <= N; i++) {

            Arrays.fill(visited, false);

            if (dfs(i, 1, N, adj)) {

                return true;

            }

        }

    }

}
```

```

    }

    return false;
}

boolean dfs(int u, int count, int N, ArrayList<ArrayList<Integer>> adj) {

    visited[u] = true;

    if (count == N) {

        return true;

    }

    for (int v : adj.get(u)) {

        if (!visited[v]) {

            if (dfs(v, count + 1, N, adj)) {

                return true;

            }

        }

    }

    visited[u] = false;

    return false;

}
}

```

Submission:

The screenshot displays a coding platform interface with a dark theme. On the left, the 'Output Window' shows 'Compilation Results' for 'Y.O.G.I. (AI Bot)'. It indicates 'Problem Solved Successfully' with a green checkmark. Below this, statistics are shown: 'Test Cases Passed: 52 / 52', 'Attempts: Correct / Total: 1 / 1', 'Accuracy: 100%', 'Points Scored: 4 / 4', and 'Your Total Score: 12' with a green upward arrow. A 'Solve Next' button is at the bottom left. On the right, the code editor shows a Java solution for a problem involving a graph and DFS. The code includes imports for ArrayList and Arrays, a class Solution, and a check method that initializes a visited array and calls a recursive dfs method. At the bottom right, there are buttons for 'Custom Input', 'Compile & Run', and 'Submit'.

```

1 // User Function Template for Java
2 import java.util.ArrayList;
3 import java.util.Arrays;
4
5 class Solution {
6
7     boolean[] visited;
8
9     boolean check(int N, int M, ArrayList<ArrayList<Integer>> Edges) {
10
11         ArrayList<ArrayList<Integer>> adj = new ArrayList<>();
12         for (int i = 0; i <= N; i++) {
13             adj.add(new ArrayList<>());
14         }
15
16         for (ArrayList<Integer> edge : Edges) {
17             int u = edge.get(0);
18             int v = edge.get(1);
19             adj.get(u).add(v);
20             adj.get(v).add(u);
21         }
22
23         this.visited = new boolean[N + 1];
24
25         for (int i = 1; i <= N; i++) {
26             Arrays.fill(visited, false);
27
28             if (dfs(i, 1, N, adj)) {
29                 return true;
30             }
31         }
32     }
33 }

```